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WIPRO NGA Program _26SUB0726-JAVA J2EE INTEGRATION

Capstone Project Presentation - 23rd June 2026

Project Title - Vehicle Service Management System

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Vehicle Service Management System

Introduction

The Vehicle Service Management System is a modern web-based application built to automate service centre operations. It provides workshops with a centralized digital platform to manage customer profiles and vehicle ownership records. By replacing manual paper logs, the system eliminates human calculation errors and improves tracking efficiency. Administrators can securely log in to the dashboard using seeded credentials protected by BCrypt password encryption.

The application handles the complete servicing lifecycle from initial booking registration to workshop mechanical repairs. Service statuses are updated dynamically in real-time, transitioning from Booked to In Progress and finally Completed. Upon service completion, the system records log details and automatically compiles itemized digital billing invoices. Built on Java and Spring Boot, the backend ensures secure endpoint authorization and so this project delivers a highly secure, scalable, and production-ready vehicle service management solution.

Project Objectives

Objectives

- **Process Automation:** Replace manual registers with automated booking allocations and workshop service tracking that can go under dynamical changes .
- **Dynamic Lifecycle Management:** Enable step-by-step progress tracking of vehicles in servicing from Booked to Completed by admin throughout.
- **Centralized Data Storage:** Establish a secure relational database schema linking customers, vehicles, bookings, logs, and invoices.
- **Automated Financial Billing:** Eliminate calculation mistakes by generating invoice values automatically based on catalog rates. It eliminates human financial calculation errors by generating itemized service invoices automatically.
- **Security :** It encrypts sensitive user credentials and administrator passwords using the BCrypt hashing algorithm.
- **Enterprise DevOps Standards:** Deploy application packages inside Docker container configurations orchestrated using Kubernetes clusters.
- **CI-CD Automation:** Automate build tasks, testing routines, and updates using a declarative Jenkins pipeline script.

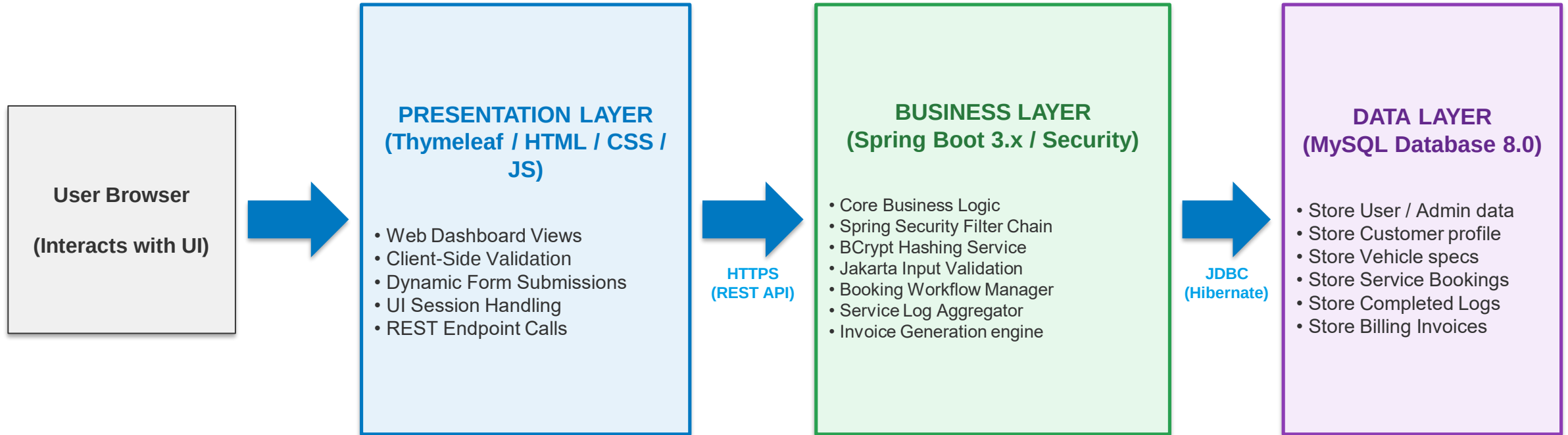
Scope of the Project

Scope

- **Secure Authentication Boundary:** Protects administrative dashboard views and REST endpoints using a Spring Security filter chain. It secures credentials by hashing all administrator passwords using the BCrypt .
- **Customer and Vehicle Profile Manager:** Manages database records for customer profiles and links them directly to vehicle specifications. It provides full CRUD operations to add, view, update, and search ownership records.
- **Dynamic Service Records:** Archives previous servicing logs, mechanic remarks, repair categories, and parts costs. It tracks the active service status dynamically as it moves from booking to completion.
- **Digital Billing Pipeline:** Automates service cost calculations using catalog base rates upon workshop completion. It generates itemized billing statements and outputs downloadable digital invoice logs.
- **Integrated REST API Suite:**Exposes complete CRUD endpoints documented with interactive Swagger OpenAPI web pages. It provides preconfigured Postman collection suites containing payloads for automated API testing.
- **High-Availability Deployment:**Packages application containers using Docker and coordinates deployments with Kubernetes manifests. It automates software builds, unit test reports, and rolling restarts using Jenkins pipelines.

System Architecture

Three-Tier Architecture Diagram



System Modules & Features

System Modules

- **Administrative Security Portal:** It is the main module that handles administrator logins secured by BCrypt password validations, preventing unauthenticated access.
- **Customer Account Directory:** This module manages profiles containing customer names, email addresses, phone contact data, and residential details.
- **Vehicle Inventory System:** This module helps to records vehicle license plate registrations, brand makes, models, and manufacture years, linking them to owner IDs.
- **Service Booking Engine:** Creates bookings containing requested service catalog items, setting status identifiers dynamically.
- **Workshop Log Manager:** Oversees mechanical servicing records, transitioning statuses from In Progress to Completed upon repair finalization.
- **Automated Invoice Engine:** Calculates pricing details and compiles downloadable bills featuring timestamp logs.

Technologies Used

Technology Stack

Frontend Technologies Used :

- **HTML5** – Used for creating the structure and layout of web pages.
- **CSS3** – Used for styling and designing the user interface.
- **JavaScript** – Used for client-side interactions and dynamic behavior.
- **Thymeleaf** – Used as the server-side template engine for rendering views and integrating frontend with backend data.

Backend Technologies Used :

- **Java 17** – Used as the core programming language and execution platform.
- **Spring Boot 3.x** – Used for developing and configuring the application.
- **Spring Security** – Used for authentication, authorization, and securing APIs.

Persistence Layer Technologies Used :

- **MySQL 8.0** – Used as the relational database for storing application data.
- **Hibernate JPA** – Used for Object-Relational Mapping (ORM) and database operations.

Server Environment :

- **Apache Tomcat (Embedded)** – Used as the web server inside the Spring Boot executable JAR.
- **Port 8086** – Used for serving the application and handling HTTP requests.

Testing Technologies :

- **JUnit 5** – Used for unit testing backend components.
- **Mockito** – Used for mocking dependencies and testing individual classes.

Security Technologies :

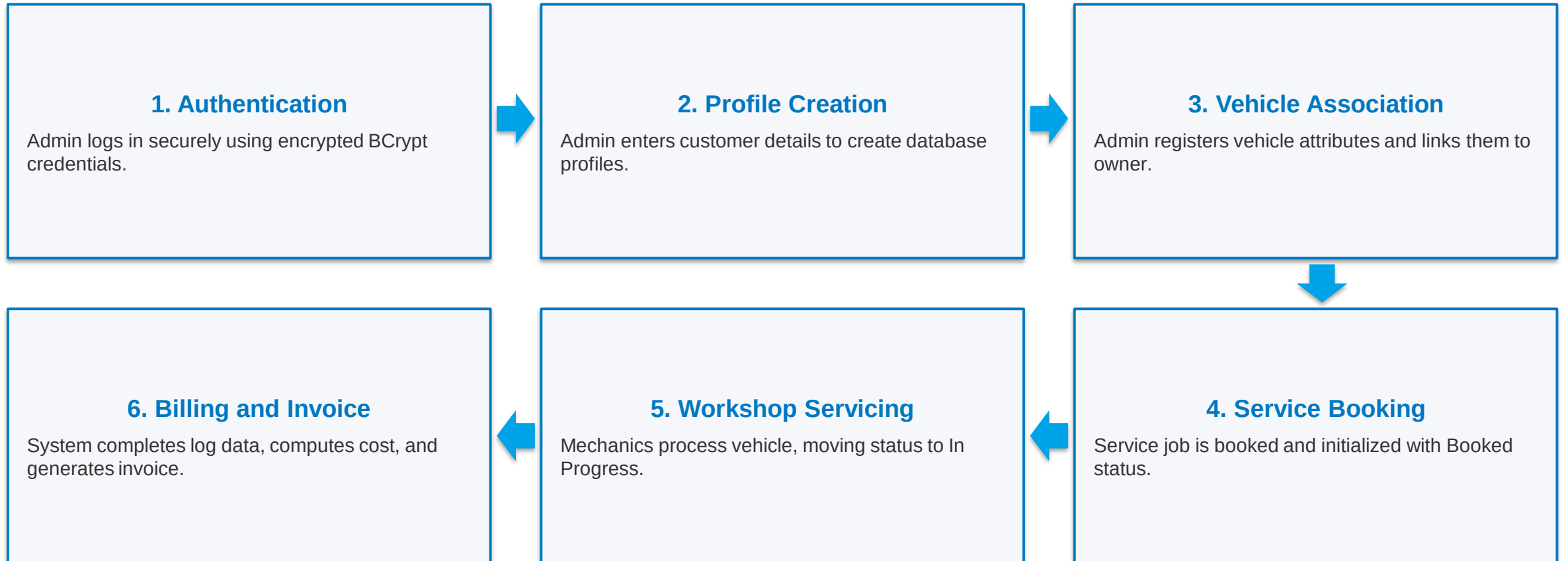
- **Spring Security** – Used for authentication and authorization.

API Documentation and Testing Tools :

- **Swagger-UI** – Used for interactive API documentation and testing endpoints.
- **Postman** – Used for sending requests and verifying API functionality programmatically.

System Workflow

Admin-Centric Operations Flow



DevOps, Deployment & Benefits

DevOps & Deployment

- **Docker Containerization:** Employs a multi-stage Dockerfile to build project code and execute applications in clean runtime environments. Packages the Spring Boot application and database services into isolated container images. This process minimizes the overall storage footprint and ensures the application runs consistently across different operating systems.
- **Docker Compose:** Coordinates the local multi-container environment by running the web application and MySQL database server side-by-side. It maps the container port 8086 to the host machine, defines database environment variables, and establishes virtual networks. This allows developers to spin up and test the entire project stack locally using a single command.
- **Kubernetes Orchestration:** Automates the deployment, networking, and scaling of application pods inside a local Minikube cluster. It utilizes manifest files to configure services and NodePorts, mounts ConfigMaps for environment variables, and secures access .
- **Jenkins CI-CD Pipeline:** Automates the software delivery workflow from code check-in to active cluster rollout. The declarative pipeline executes stages for checking out code, running unit tests, generating Surefire reports, and building tagged Docker images. It then logs into the image registry, pushes the package, and executes a rolling rollout restart on Kubernetes.

Testing & Quality Assurance

Verification and Testing Suite

- **Unit & Mock Testing**

- Uses JUnit 5 and Mockito to verify the backend business logic.
- Tests customer registration rules and checks for duplicate data entries.
- Verifies that booking status transitions occur correctly from booking to completion.

- **Security & Validation Tests**

- Uses MockMvc to test admin security controls and login routing rules.
- Confirms that unauthenticated traffic is blocked and redirected to the login page.
- Validates that Jakarta annotations reject incorrect input values at controller boundaries.

- **API Testing & Verification**

- Exposes interactive Swagger UI pages to test REST endpoints visually in a browser.
- Provides a complete Postman collection containing CRUD requests for all features.
- Automates request verification, checking cookies, headers, and validation error responses.

Benefits & Drawbacks

Project Evaluation

BENEFITS:

- End-to-End Automation: Replaces paper logs, saving mechanical and booking schedules.
- CI-CD Deployment Reliability: Integrates Jenkins to run test jobs automatically.
- Strong Cryptographic Security: Protects passwords using BCrypt encryption.
- System Portability: Dockerizes application components to run in any Kubernetes host.

DRAWBACKS & LIMITATIONS:

- Database Single Point of Failure: The database runs in a single pod configuration rather than a multi-node replica set.
- Admin-Only Workspace Dashboard: The UI does not provide customer-facing access profiles.
- High Execution Footprints: Requires hosting Docker, Minikube, and Jenkins locally, consuming high resources.
- Lacks Notification Channels: The system lacks active email alert channels or SMS confirmation alerts.

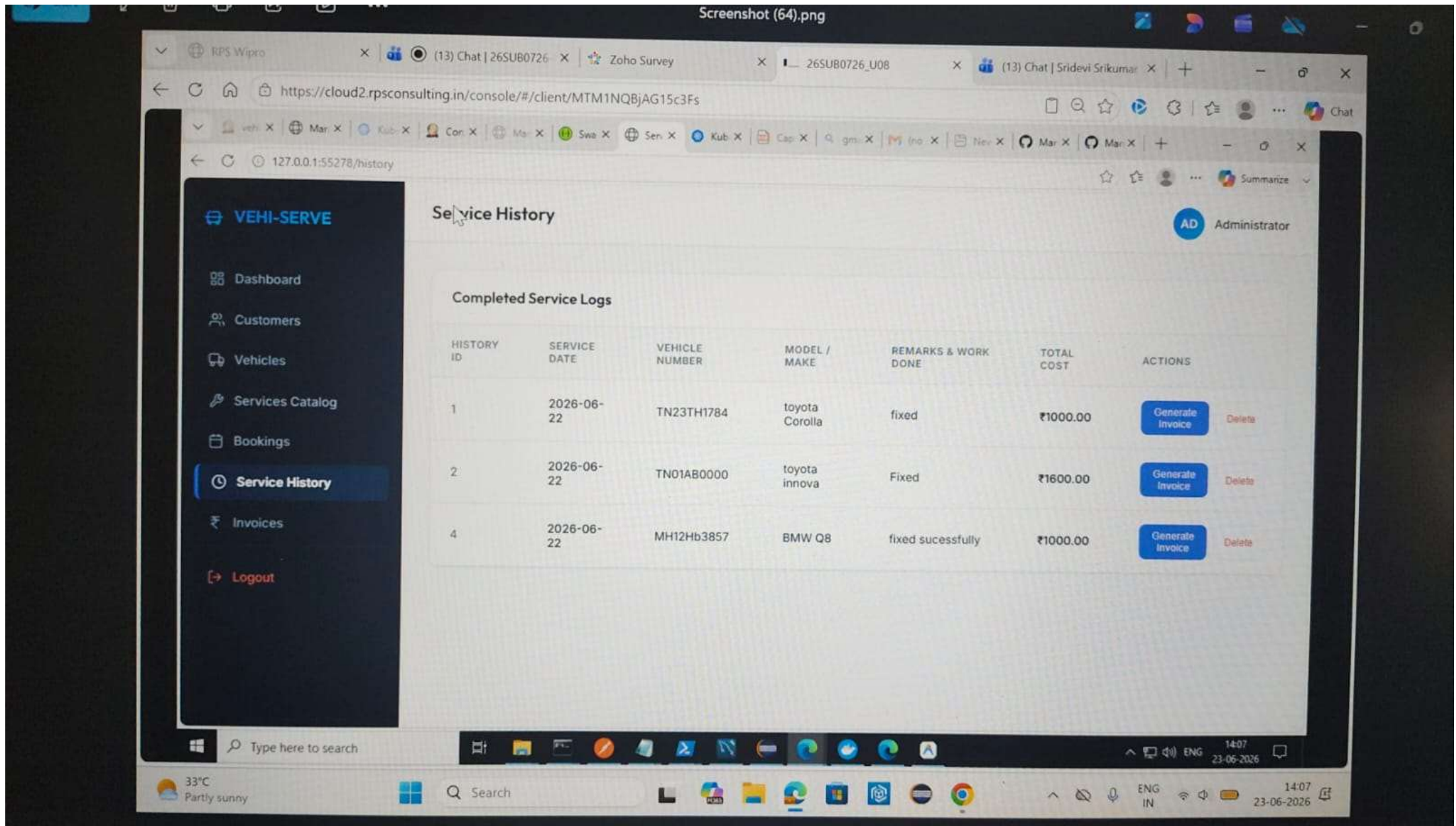
Vehicle Service Management System

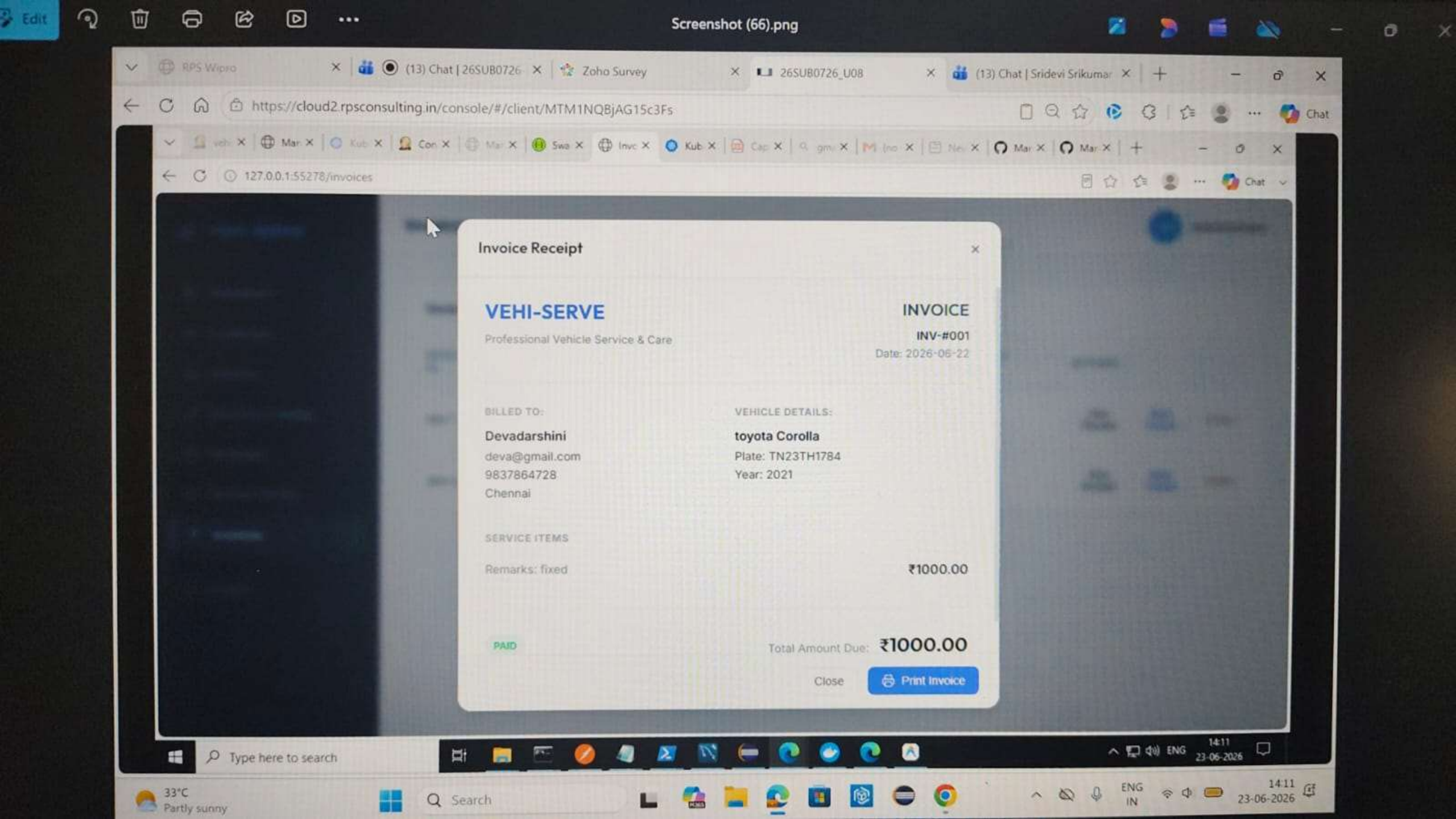
Conclusion

The Vehicle Service Management System successfully delivers a secure, automated, and production-ready platform for workshop operations. By integrating a Spring Boot backend, a Thymeleaf frontend, and a relational MySQL database, the system streamlines customer management, booking lifecycles, and billing invoices. The project achieves modern deployment standards through Docker containerization, Kubernetes cluster orchestration, and Jenkins CI/CD pipeline automation. Overall, this capstone project demonstrates a complete software engineering lifecycle with robust security configurations, automated testing coverage, and cloud-native infrastructure readiness.

Future Enhancements:

- Develop direct client-facing booking interfaces.
- Alert Notifications such as Send automatic SMS alerts, Email confirmation receipts, Remind about upcoming services.
- Implement automated daily backups to secure cloud storage repositories.
- Enable clients to track their vehicle's active repair status and view previous service histories.
- Allow customers to pay their service invoices digitally directly from the application.





Browser tabs: RPS Wipro, (13) Chat | 26SUB0726, Zoho Survey, 26SUB0726_U08, (13) Chat | Sridevi Srikumar

Address bar: <https://cloud2.rpsconsulting.in/console/#/client/MTM1NQ8jAG15c3Fs>

Browser tabs (inner): web X, Mar X, Kub X, Con X, Mar X, Swa X, Log X, Kub X, Cap X, gmi X, Inc X, Nes X, Mar X, Mar X

Address bar (inner): 127.0.0.1:55278/login

VEHI-SERVE
Administrative Control Panel

Username

Password

Sign In

Windows taskbar: Type here to search, 33°C Partly sunny, Search, Taskbar icons (File Explorer, Edge, etc.), System tray: ENG, 14:05, 23-06-2026

Browser tabs: RPS Wipro, (13) Chat | 26SUB0726, Zoho Survey, 26SUB0726_U08, (13) Chat | Sridevi Srikumar

Address bar: <https://cloud2.rpsconsulting.in/console/#/client/MTM1NQBJAG15c3Fs>

Browser tabs: veh, Mar, Kub, Con, Mar, Swa, Mar, Kub, Cap, gms, Ino, Nei, Mar, Mar

Address bar: localhost:8080/job/vehiserve-pipeline/4/consoleFull

Jenkins | vehiserve-pipeline | #4

```
C:\Users\Admin\.jenkins\workspace\vehiserve-pipeline>kubectl rollout restart deployment/vehiserve-app-deployment
deployment.apps/vehiserve-app-deployment restarted
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Pipeline completed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
[Checks API] No suitable checks publisher found.
Finished: SUCCESS
```

Jenkins 2.555.2

Windows taskbar: Type here to search, 33°C Partly sunny, Search, 14:04 23-06-2026

Browser tabs: RPS Wipro, (13) Chat | 26SUB0726-Live S..., 26SUB0726_U08, Mail - Devadarshini - Outlook

Address bar: <https://cloud2.rpsconsulting.in/console/#/client/MTM1NQBJAG15c3Fs>

Sub-address bar: 127.0.0.1:55418/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/#/shell/default/mysql-deployment-85c8c578cc-mm57s/m...

kubernetes default Search

Workloads > Pods > mysql-deployment-85c8c578cc-mm57s > Shell

Workloads

- Cron Jobs
- Daemon Sets
- Deployments
- Jobs
- Pods
- Replica Sets
- Replication Controllers
- Stateful Sets
- Service
- Ingresses
- Ingress Classes
- Services
- Config and Storage

Shell in mysql in mysql-deployment-85c8c578cc-mm57s

```
| vehicle_service |
+-----+
5 rows in set (0.17 sec)

mysql> use vehicle_service;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SELECT * FROM vehicle_service.customer;
+-----+-----+-----+-----+-----+
| customer_id | address | email | name | phone |
+-----+-----+-----+-----+-----+
| 1 | Chennai | deva@gmail.com | Devadarshini | 9837864728 |
| 2 | chennai | sri@gmail.com | Sridevi | 8976456473 |
| 6 | mumbai | sam@gmail.com | sam | 9876543210 |
| 8 | hyderabad | aksh@gmail.com | Akshya | 8492784094 |
| 11 | neyveli | sri@gmail.com | sri | 9876543212 |
| 12 | delhi | dhsn@gmail.com | dm | 5874963973 |
| 13 | delhi | dhsn@gmail.com | dm | 5874963973 |
+-----+-----+-----+-----+-----+
7 rows in set (0.02 sec)

mysql>
```

Windows taskbar: Type here to search, 31°C Mostly cloudy, Search, 11:58 23-06-2026

VEHI-SERVE

Dashboard

Customers

Vehicles

Services Catalog

Bookings

Service History

Invoices

Logout

Dashboard

TOTAL CUSTOMERS

8

VEHICLES SERVICED

3

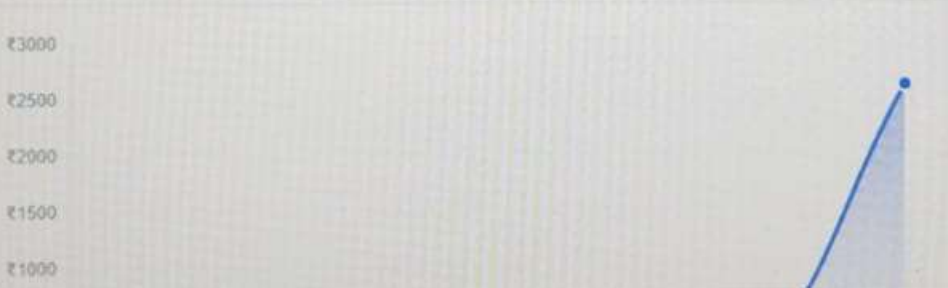
ACTIVE BOOKINGS

3

TOTAL REVENUE

₹2,600.00

Monthly Revenue Trends



Quick Actions

+ New Booking

Add New Customer

View Billing Overview

AD Administrator

Browser tabs: RPS Wipro, (13) Chat | 26SUB0726, Zoho Survey, 26SUB0726_U08, (13) Chat | Sridevi Srikumar

Address bar: <https://cloud2.rpsconsulting.in/console/#/client/MTM1NQBJAG15c3Fs>

Browser tabs: veh, Mar, Kut, Con, Ma, Swa, Mar, Kub, Cap, gm, no, New, Mar, Mar

Address bar: 127.0.0.1:55278/vehicles

VEHI-SERVE

Administrator

- Dashboard
- Customers
- Vehicles**
- Services Catalog
- Bookings
- Service History
- Invoices
- Logout

Vehicles

Vehicles Registry

[+ Add Vehicle](#)

ID	PLATE NUMBER	BRAND / MAKE	MODEL	YEAR	OWNER (CUSTOMER)	ACTIONS
1	TN23TH1784	toyota	Corolla	2021	Devadarshini	Edit Delete
2	TN01AB0000	toyota	innova	1900	sam	Edit Delete
3	MH12Hb3857	BMW	Q8	2022	Akshya	Edit Delete

Windows taskbar: 33°C Partly sunny, Search, 14:07 23-06-2026

Browser tabs: RPS Wipro, (13) Chat | 26SUB0726, Zoho Survey, 26SUB0726_U08, (13) Chat | Sridevi Srikumar

Address bar: <https://cloud2.rpsconsulting.in/console/#/client/MTM1NQBJAG15c3Fs>

Sub-address bar: 127.0.0.1:55278/bookings

VEHI-SERVE

- Dashboard
- Customers
- Vehicles
- Services Catalog
- Bookings**
- Service History
- Invoices
- Logout

Bookings

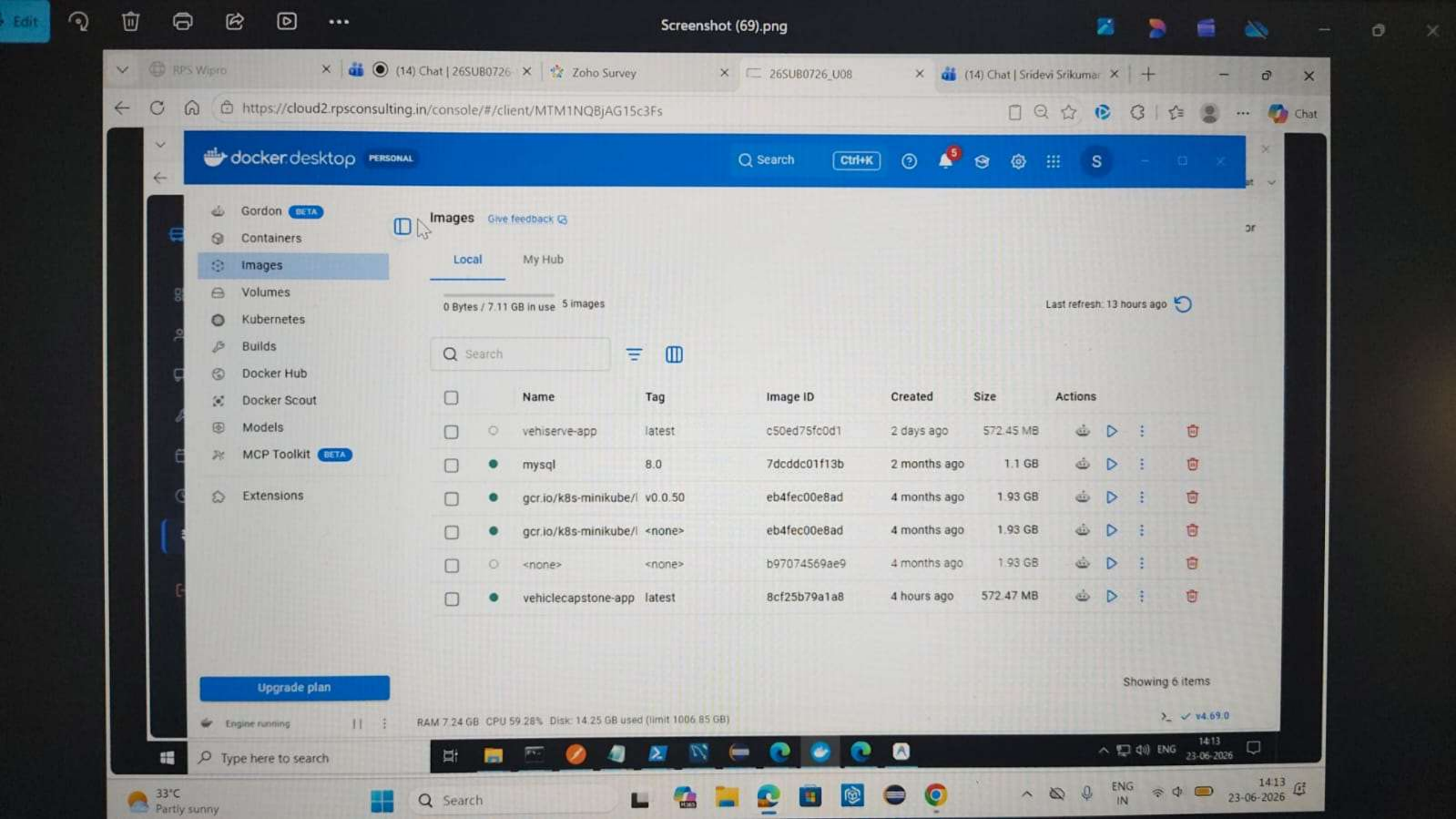
AD Administrator

Service Bookings Log

[+ New Booking](#)

ID	VEHICLE (PLATE)	OWNER	SELECTED SERVICE	PROBLEM DESCRIPTION	STATUS	ACTIONS
1	toyota Corolla (TN23TH1784)	Devadarshini	(₹)	Brakes need to be fixed	COMPLETED	Edit Delete
2	toyota innova (TN01AB0000)	sam	(₹)	General water wash	COMPLETED	Edit Delete
4	BMW Q8 (MH12Hb3857)	Akshya	(₹)	issue in brake	COMPLETED	Edit Delete

Windows taskbar: 33°C Partly sunny, Search, 14:07 23-06-2026



Gordon BETA

Containers

Images

Volumes

Kubernetes

Builds

Docker Hub

Docker Scout

Models

MCP Toolkit BETA

Extensions

Images Give feedback

Local

My Hub

0 Bytes / 7.11 GB in use 5 images

Last refresh: 13 hours ago

Search

	Name	Tag	Image ID	Created	Size	Actions
☐	vehiserve-app	latest	c50ed75fc0d1	2 days ago	572.45 MB	
☐	mysql	8.0	7dcddc01f13b	2 months ago	1.1 GB	
☐	gcr.io/k8s-minikube/l	v0.0.50	eb4fec00e8ad	4 months ago	1.93 GB	
☐	gcr.io/k8s-minikube/l	<none>	eb4fec00e8ad	4 months ago	1.93 GB	
☐	<none>	<none>	b97074569ae9	4 months ago	1.93 GB	
☐	vehiclecapstone-app	latest	8cf25b79a1a8	4 hours ago	572.47 MB	

Upgrade plan

Showing 6 items

Engine running

RAM 7.24 GB CPU 59.28% Disk: 14.25 GB used (limit 1006.85 GB)

v4.69.0

Type here to search

Search

33°C Partly sunny

14:13 23-06-2026

14:13 23-06-2026

VEHI-SERVE

Dashboard

Customers

Vehicles

Services Catalog

Bookings

Service History

₹ Invoices

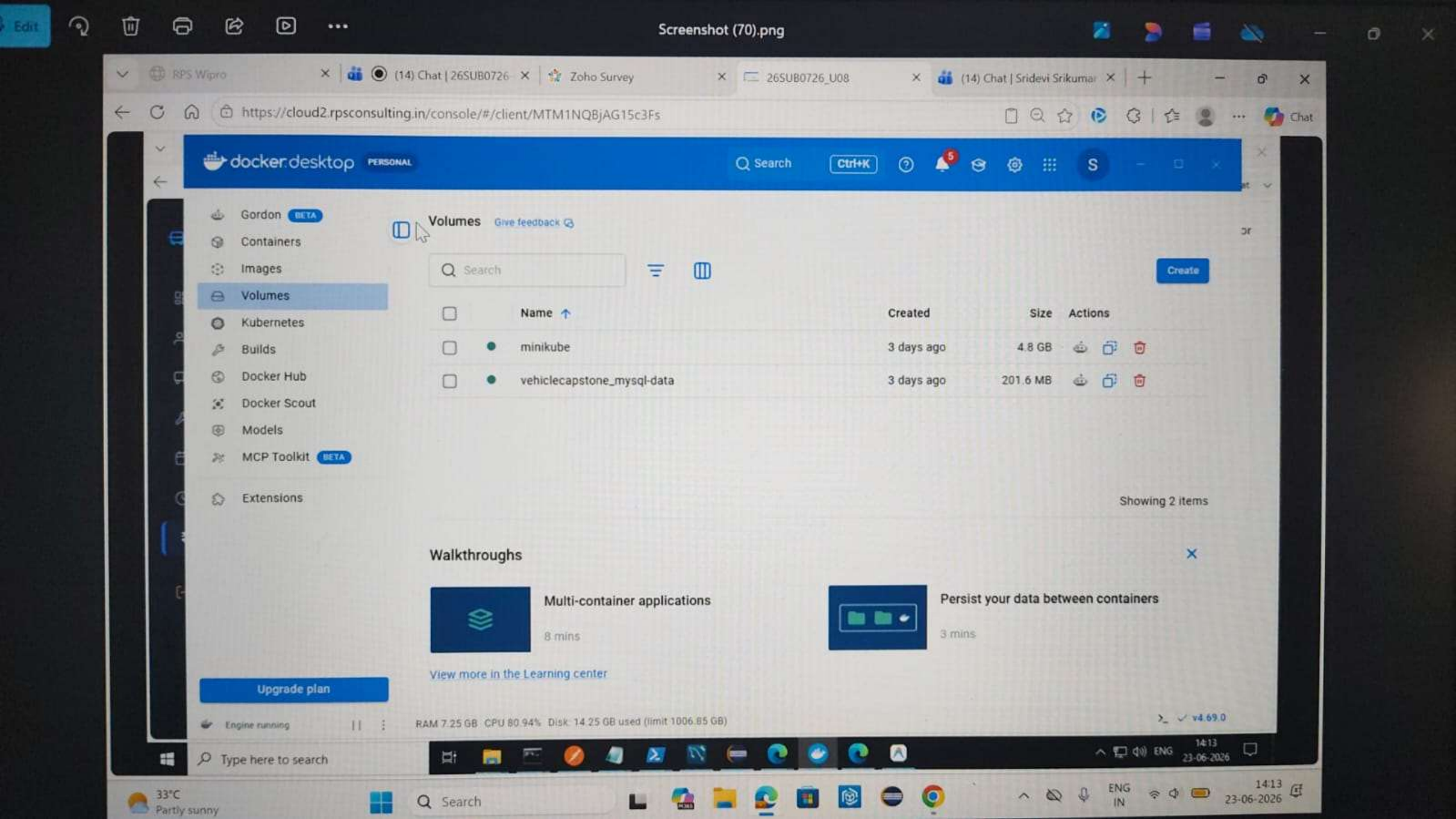
Logout

Invoices

AD Administrator

Invoices Directory

INVOICE ID	DATE	CUSTOMER NAME	VEHICLE NUMBER	AMOUNT	PAYMENT STATUS	ACTIONS
INV-1	2026-06-22	Devadarshini	TN23TH1784	₹1000.00	PAID	View Receipt Mark Unpaid Delete
INV-2	2026-06-22	sam	TN01AB0000	₹1600.00	PAID	View Receipt Mark Unpaid Delete



- Gordon BETA
- Containers
- Images
- Volumes
- Kubernetes
- Builds
- Docker Hub
- Docker Scout
- Models
- MCP Toolkit BETA
- Extensions

Volumes Give feedback

Search

	Name	Created	Size	Actions
	minikube	3 days ago	4.8 GB	
	vehiclecapstone_mysql-data	3 days ago	201.6 MB	

Walkthroughs



Multi-container applications

8 mins



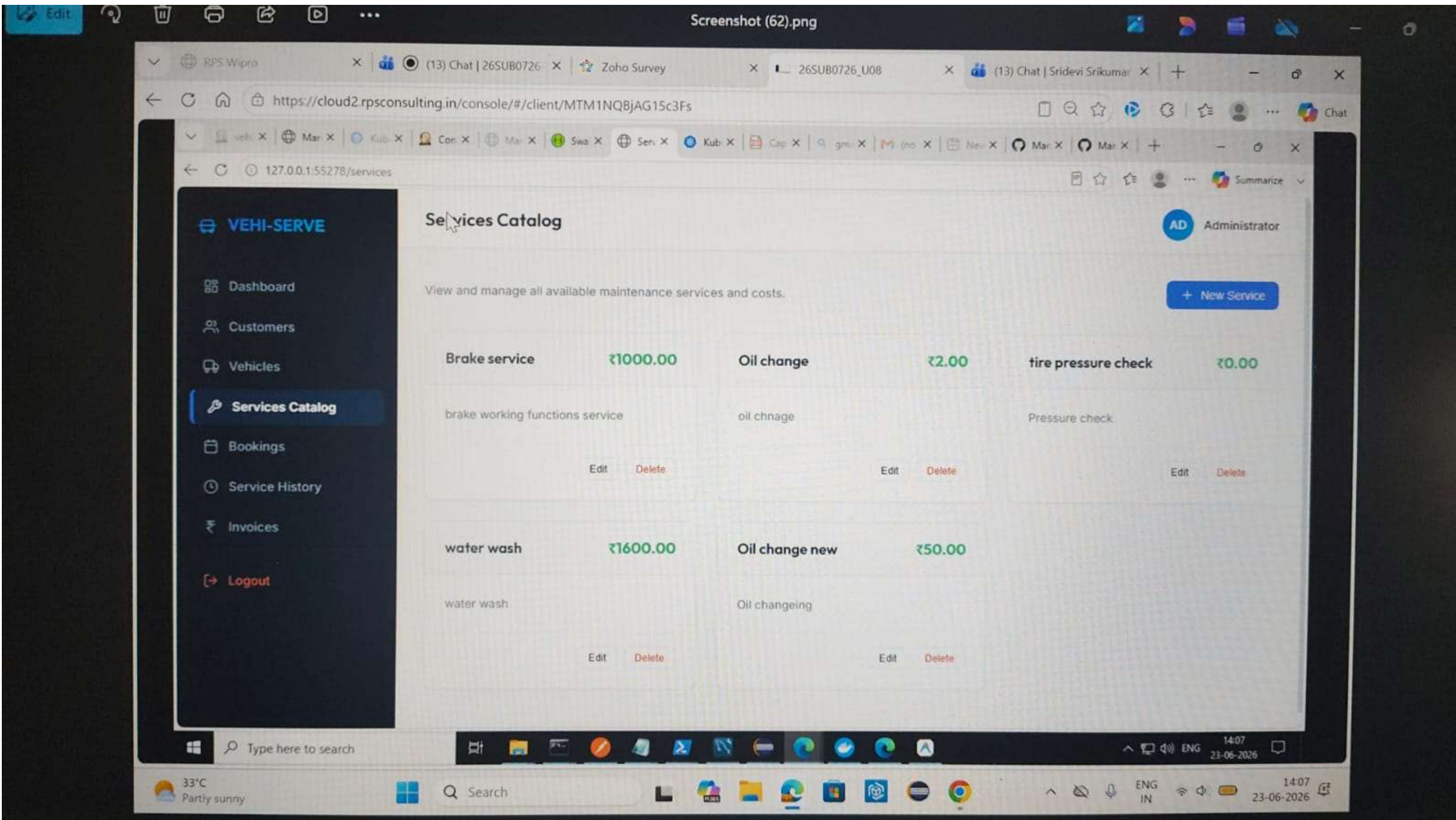
Persist your data between containers

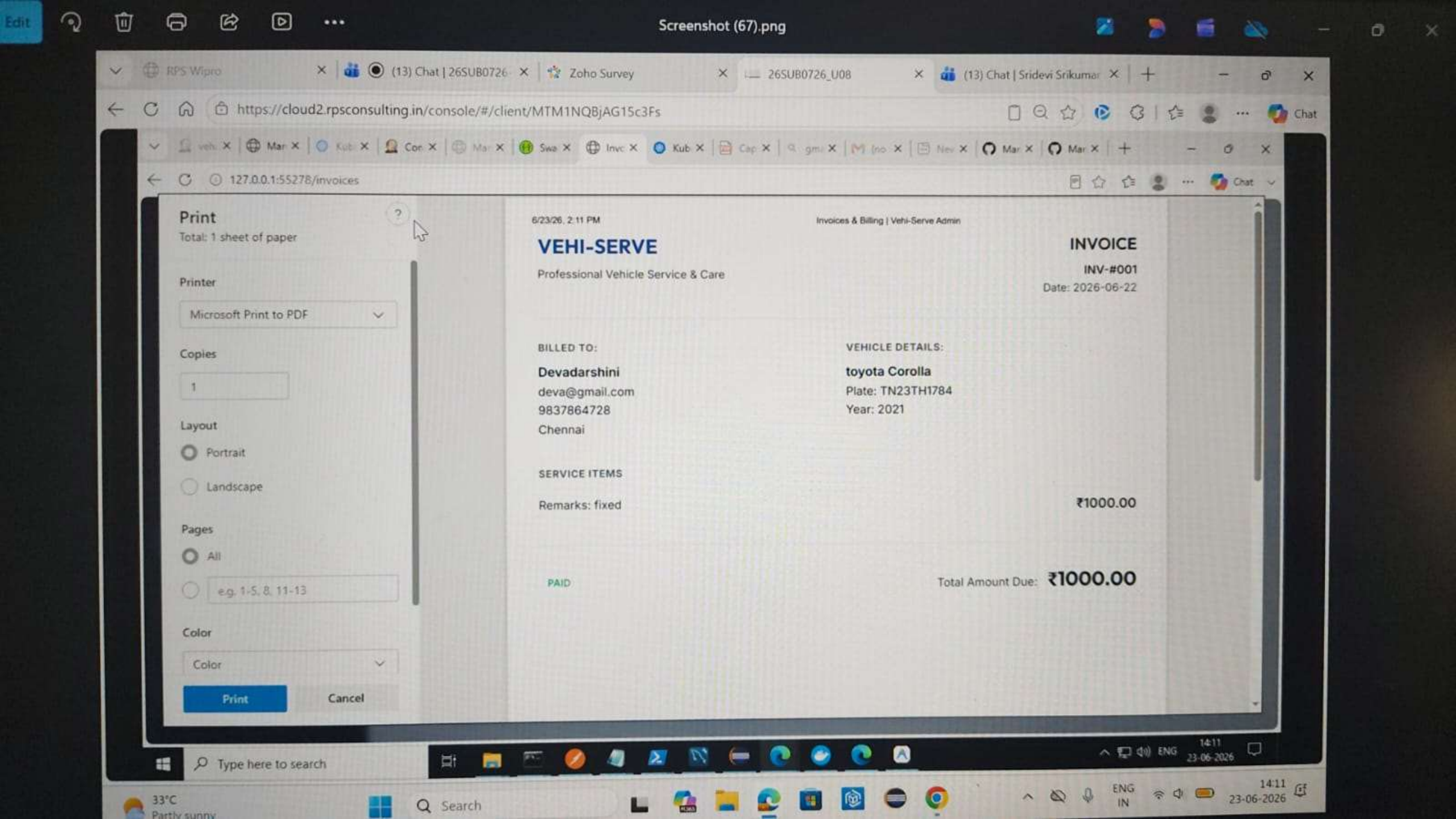
3 mins

[View more in the Learning center](#)

Upgrade plan

Engine running RAM 7.25 GB CPU 80.94% Disk 14.25 GB used (limit 1006.85 GB) v4.69.0





VEHI-SERVE sidebar menu with navigation items: Dashboard, Customers, Vehicles, Services Catalog, Bookings, Service History, Invoices, and Logout.

Customers

AD Administrator

Customers Directory

+ New Customer

ID	NAME	EMAIL ADDRESS	PHONE NUMBER	ADDRESS	ACTIONS
1	Devadarshini	deva@gmail.com	9837864728	Chennai	Vehicles Edit Delete
2	Sridevi	sri@gmail.com	8976456473	chennai	Vehicles Edit Delete
6	sam	sam@gmail.com	9876543210	mumbai	Vehicles Edit Delete
8	Akshya	aksh@gmail.com	8492784094	hyderabad	Vehicles Edit Delete
11	sri	sri@gmail.com	9876543212	neyveli	Vehicles Edit Delete
12	dmv	dhsn@gmail.com	5874963973	delhi	Vehicles Edit Delete
13	dmv	dhsn@gmail.com	5874963973	delhi	Vehicles Edit Delete

